CSE 312

Foundations of Computing II

Lecture 3: Even more counting

Binomial Theorem, Inclusion-Exclusion, Pigeonhole Principle

Recap

- Two core rules for counting a set S:

  - Sum rule:
    - Break up S into disjoint pieces/cases
    - $|S|$ = the sum of the sizes of the pieces.

  - Product rule:
    - View the elements of S as being constructed by a series of choices, where the # of possibilities for each choice doesn't depend on the previous choices
    - $|S|$ = the product of the # of choices in each step of the series.

Recap

- k-sequences: How many length k sequences over alphabet of size n?
  - Product rule ➔ n^k
- k-permutations: How many length k sequences over alphabet of size n, without repetition?
  - Permutation ➔ $\frac{n!}{(n-k)!}$
- k-combinations: How many size k subsets of a set of size n (without repetition and without order)?
  - Combination ➔ $\binom{n}{k} = \frac{n!}{k!(n-k)!}$

Binomial Coefficients – Many interesting and useful properties

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!} \qquad \binom{n}{n} = 1 \qquad \binom{n}{1} = n \qquad \binom{n}{0} = 1$$

Fact. $\binom{n}{k} = \binom{n}{n-k}$ **Symmetry in Binomial Coefficients**

Fact. $\sum_{k=0}^{n}\binom{n}{k} = 2^n$ **Follows from Binomial Theorem**

Fact. $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ **Pascal's Identity**

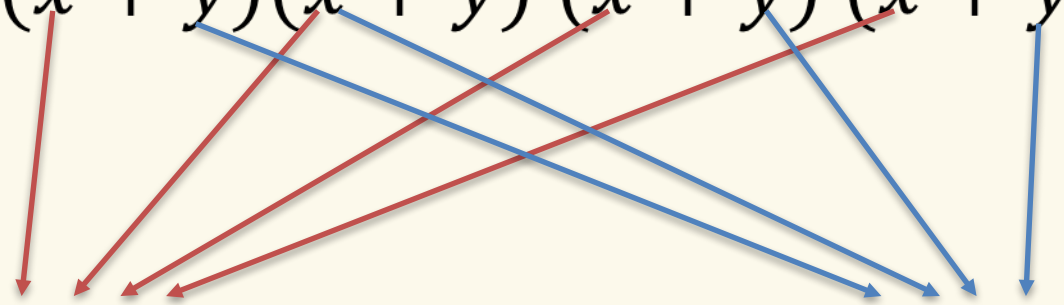
Binomial Theorem: Idea

$$(x+y)^2 = (x+y)(x+y)$$
$$= xx + xy + yx + yy$$
$$= x^2 + 2xy + y^2$$

Poll: What is the coefficient for xy^3?

A. 4 ✓
B. $\binom{4}{1}$ ✓
C. $\binom{4}{3}$ ✓
D. 3

$$(x+y)^4 = (x+y)(x+y)\,(x+y)\,(x+y)$$



$$= xxxx + yyyy + xyxy + yxyy + \ \ldots$$

Binomial Theorem: Idea

$$(x+y)^n = (x+y)\ldots(x+y)$$

Each term is of the form $x^k y^{n-k}$, since each term is made by multiplying exactly n variables, either x or y, one from each copy of $(x+y)$

How many times do we get $x^k y^{n-k}$?

The number of ways to choose x from exactly k of the n copies of $(x+y)$ (the other $n-k$ choices will be y) which is:

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$$\binom{n}{k} = \binom{n}{n-k}$$

Binomial Theorem

Theorem. Let $x, y \in \mathbb{R}$ and $n \in \mathbb{N}$ a positive integer. Then,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

Binomial Theorem

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Many properties of sums of binomial coefficients can be found by plugging in different values of x and y in the Binomial Theorem.

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Apply with $x = y = 1$

Many properties of sums of binomial coefficients can be found by plugging in different values of x and y in the Binomial Theorem.

Corollary.

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

Agenda

- Binomial Theorem
- Combinatorial Proofs
- Inclusion-Exclusion
- Pigeonhole Principle
- Counting Practice

Recall: Symmetry in Binomial Coefficients

Fact. $\binom{n}{k} = \binom{n}{n-k}$

Two equivalent ways to choose k out of n objects (unordered)

1. Choose which k elements are included
2. Choose which $n - k$ elements are excluded

Format for a combinatorial argument/proof of $a = b$

- Let S be a set of objects
- Show how to count $|S|$ one way $\Rightarrow |S| = a$
- Show how to count $|S|$ another way $\Rightarrow |S| = b$

Combinatorial argument/proof

- Elegant
- Simple
- Intuitive



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Algebraic argument

- Brute force
- Less Intuitive



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Pascal's Identity

Fact. $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$

How to prove Pascal's identity?

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How to prove Pascal's identity?

Algebraic argument:

$$\binom{n-1}{k-1} + \binom{n-1}{k} = \frac{(n-1)!}{(k-1)!\,(n-k)!} + \frac{(n-1)!}{k!\,(n-1-k)!}$$

$$= 20\ years\ later\ \ldots$$

$$= \frac{n!}{k!\,(n-k)!}$$

$$= \binom{n}{k}$$

Hard work and not intuitive

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Let's see a combinatorial argument

Example – Pascal's Identity

Fact. $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$

Combinatorial proof idea:

- Find *disjoint* sets A and B such that A, B, and $S = A \cup B$ have the sizes above.
- The equation then follows by the Sum Rule.

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$|S|$ $|A|$ $|B|$

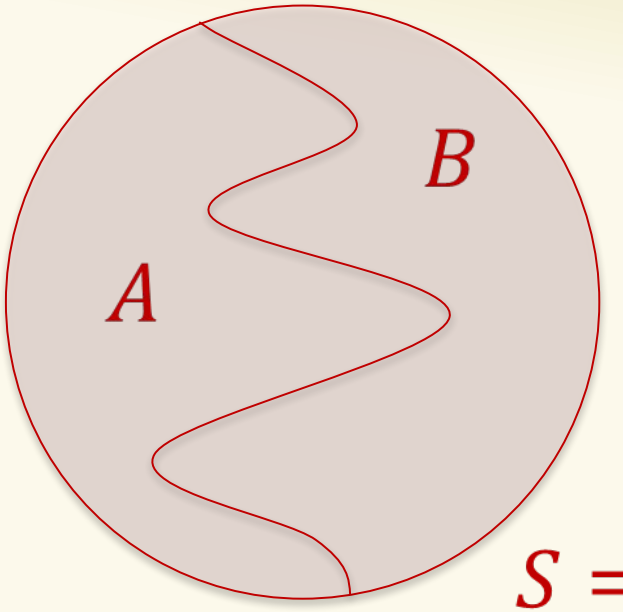
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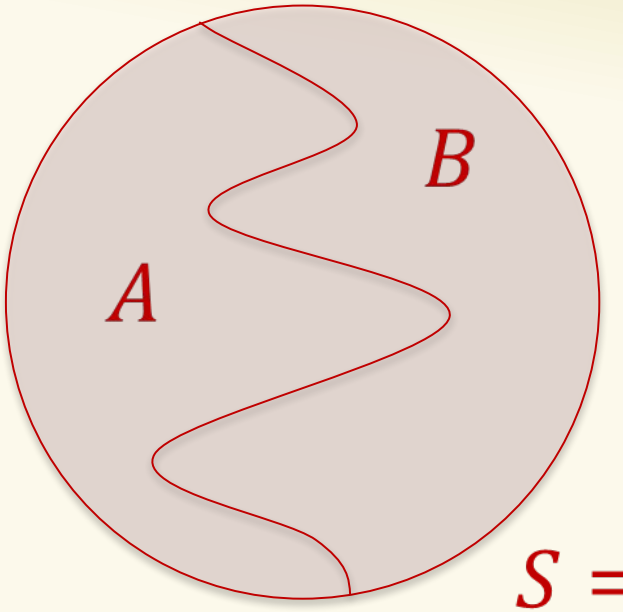
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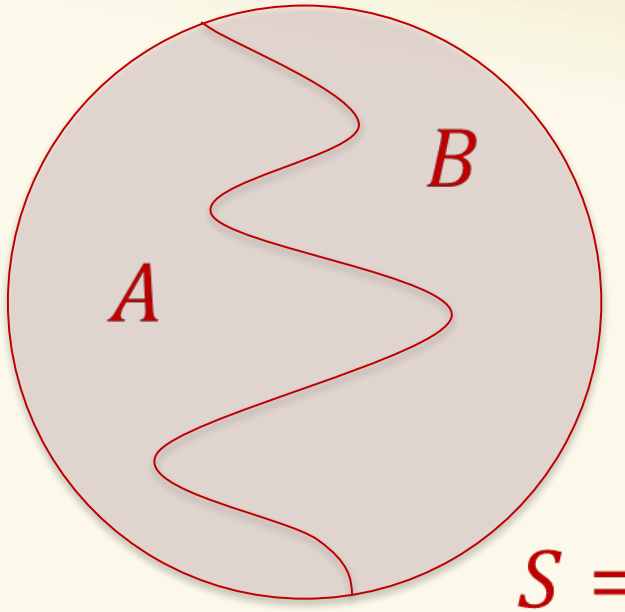
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$|S| = |A| + |B|$



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n not in set, need to choose k elements from $[n-1]$

$$|B| = \binom{n-1}{k}$$

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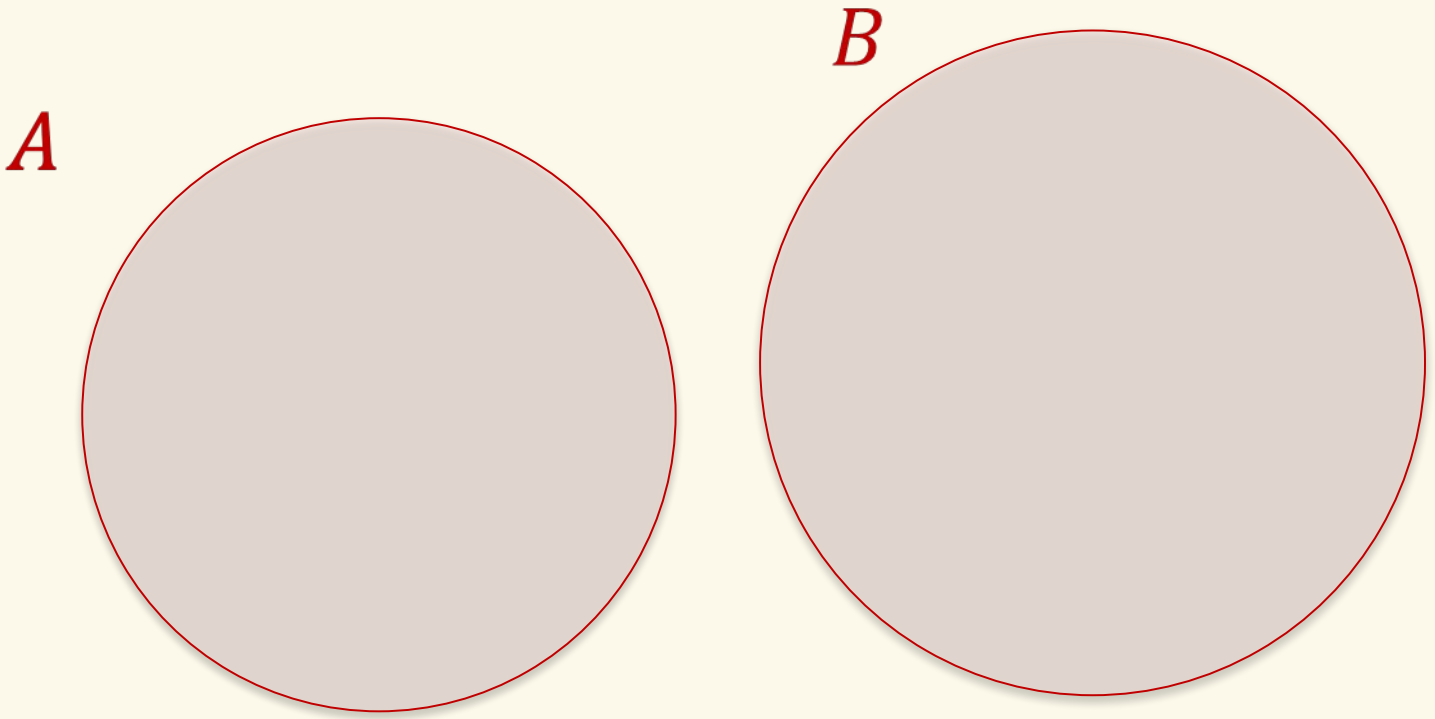
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Agenda

- Binomial Theorem
- Combinatorial Proofs
- Inclusion-Exclusion
- Pigeonhole Principle
- Counting Practice

Recap Disjoint Sets

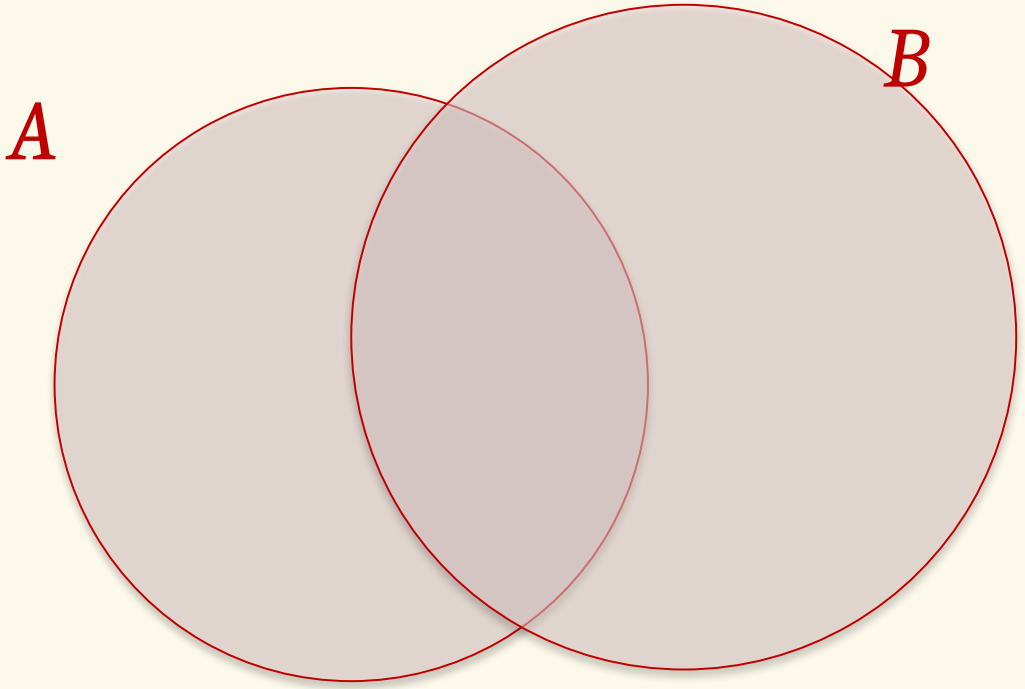
Sets that do not contain common elements ($A \cap B = \emptyset$)



Sum Rule: $|A \cup B| = |A| + |B|$

Inclusion-Exclusion

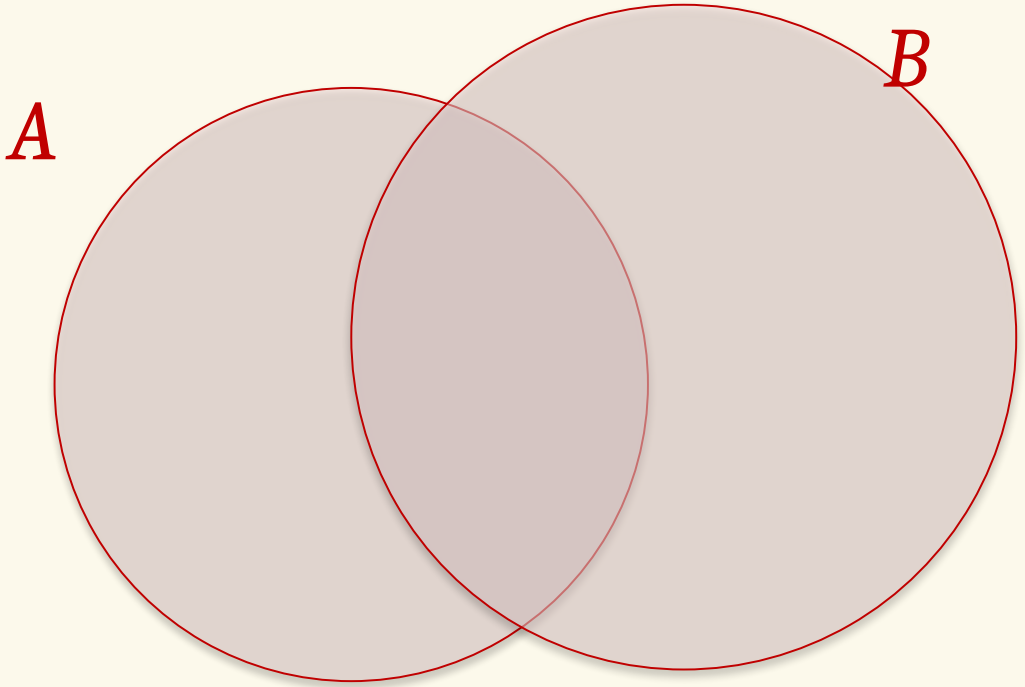
But what if the sets are not disjoint?



$|A|=43$
$|B|=20$
$|A \cap B|=7$
$|A \cup B|=$???

Inclusion-Exclusion

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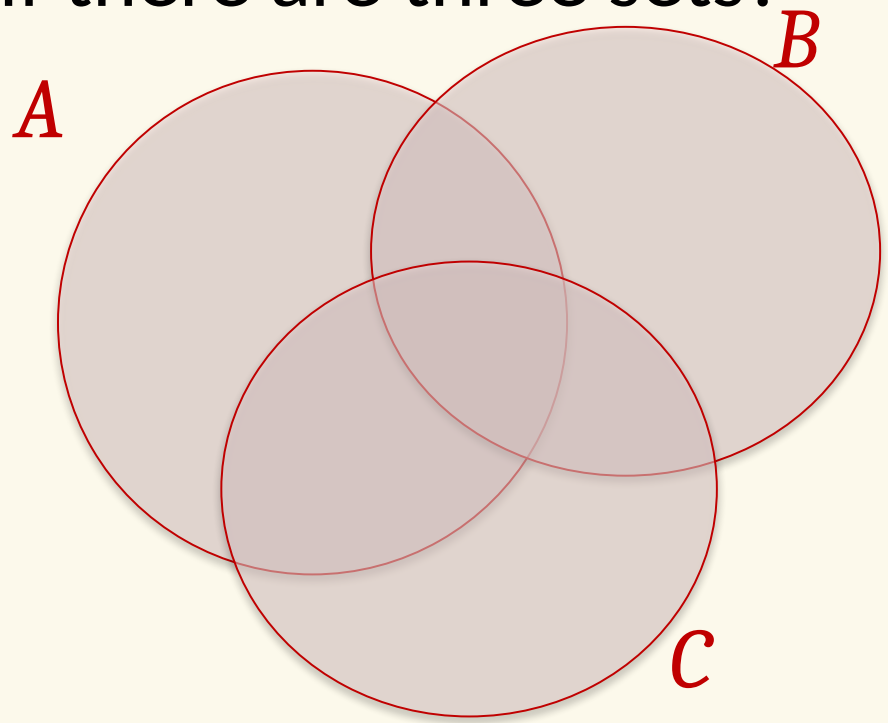


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Fact. $|A \cup B| = |A| + |B| - |A \cap B|$

Inclusion-Exclusion

What if there are three sets?



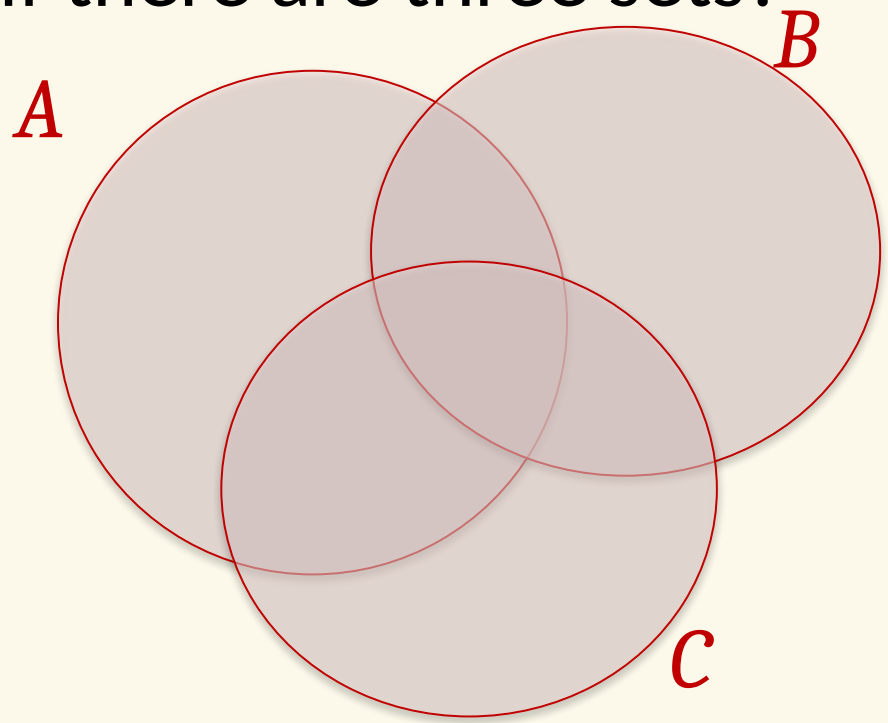
Not drawn to scale

$|A|=43$
$|B|=20$ $|C|=35$
$|A \cap B|=7$
$|A \cap C|=16$
$|B \cap C|=11$
$|A \cap B \cap C|=4$
$|A \cup B \cup C|=$???

Inclusion-Exclusion

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Fact.

$$|A\cup B\cup C|=|A|+|B|+|C| - |A\cap B| - |A\cap C|-|B\cap C|+|A\cap B\cap C|$$

Inclusion-Exclusion

Let A, B be sets. Then

$$|A \cup B| = |A| + |B| - |A \cap B|$$

In general, if $A_1, A_2, \ldots, A_n$ are sets, then

$$|A_1 \cup A_2 \cup \cdots \cup A_n| = singles - doubles + triples - quads + \ldots$$
$$= (|A_1| + \cdots + |A_n|) - (|A_1 \cap A_2| + \ldots + |A_{n-1} \cap A_n|) + \ldots$$

Brain Break



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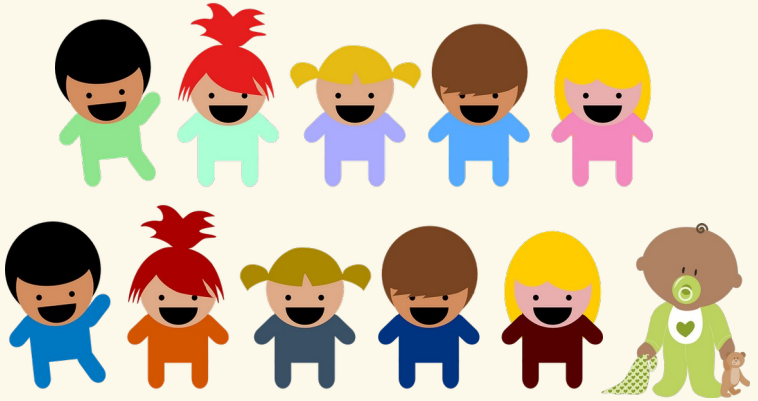
Pigeonhole Principle (PHP): Idea

10 pigeons, 9 pigeonholes



Pigeonhole Principle: Idea





If 11 children have to share 3 cakes, at least one cake must be shared by how many children?

Pigeonhole Principle – More generally

If there are n pigeons in $k < n$ holes, then one hole must contain at least $\frac{n}{k}$ pigeons!

Proof. Assume there are $< \frac{n}{k}$ pigeons per hole.

Then, there are $< k \cdot \frac{n}{k} = n$ pigeons overall.

Contradiction!

Pigeonhole Principle – Better version

- If there are n pigeons in $k < n$ holes, then one hole must contain at least $\left\lceil \frac{n}{k} \right\rceil$ pigeons!

Reason. Can't have fractional number of pigeons

Syntax reminder:

- Ceiling: $\lceil x \rceil$ is x rounded up to the nearest integer (e.g., $\lceil 2.731 \rceil = 3$)
- Floor: $\lfloor x \rfloor$ is x rounded down to the nearest integer (e.g., $\lfloor 2.731 \rfloor = 2$)

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Pigeonhole Principle: Strategy

To use the PHP to solve a problem, there are generally 4 steps

1. Identify pigeons
2. Identify pigeonholes
3. Specify how pigeons are assigned to pigeonholes
4. Apply PHP

Pigeonhole Principle – Example

In a room with 367 people, there are at least two with the same birthday.

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Solution:

1. **367** pigeons = people
2. **366** holes (365 for a normal year + Feb 29) = possible birthdays
3. Person goes into hole corresponding to own birthday
4. By PHP, there must be two people with the same birthday

Pigeonhole Principle – Example (Surprising?)

In every set S of 100 integers, there are at least **two** *elements whose difference is a multiple of 37.*

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Pigeons: integers x in S

Pigeonholes: $\{0,1,\ldots,36\}$

Assignment: x goes to $x \bmod 37$

Since $100 > 37$, by PHP, there are $x \neq y \in S$ s.t.
$x \bmod 37 = y \bmod 37$ which implies
$x - y = 37\,k$ for some integer k

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- Pigeonhole Principle
- Counting Practice

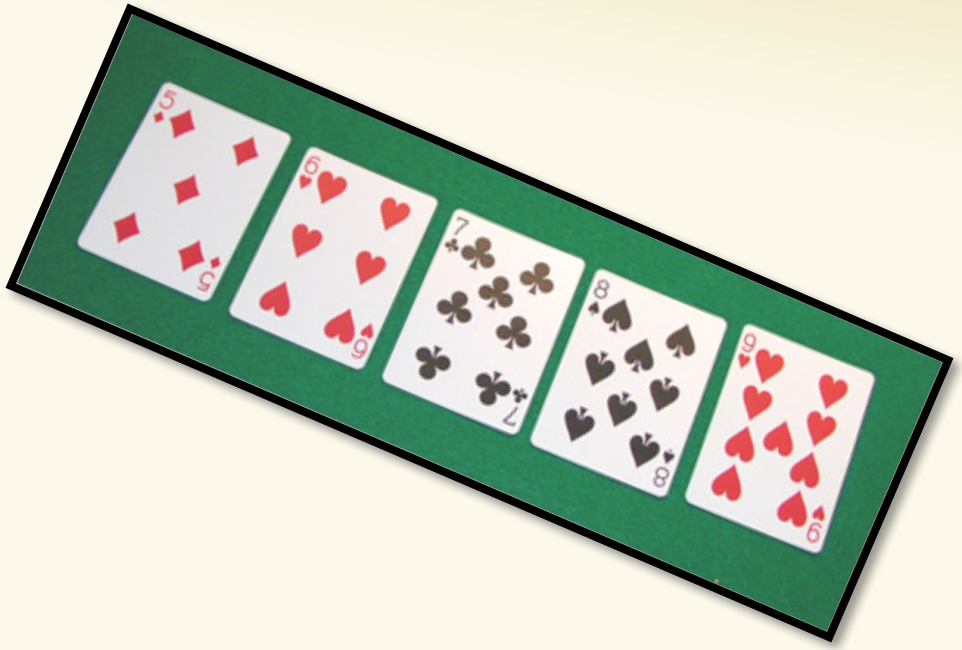
Quick Review of Cards



- 52 total cards
- 13 different ranks: 2,3,4,5,6,7,8,9,10,J,Q,K,A
- 4 different suits: Hearts, Diamonds, Clubs, Spades

Quick Review of Cards



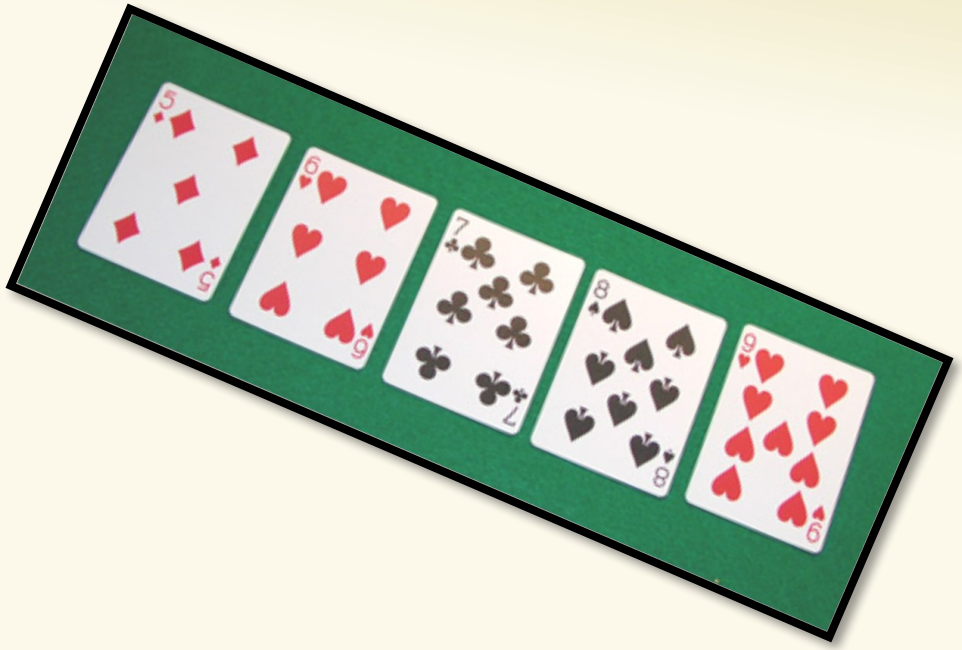


How many possible 5 card hands?

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Quick Review of Cards





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$$\binom{52}{5}$$

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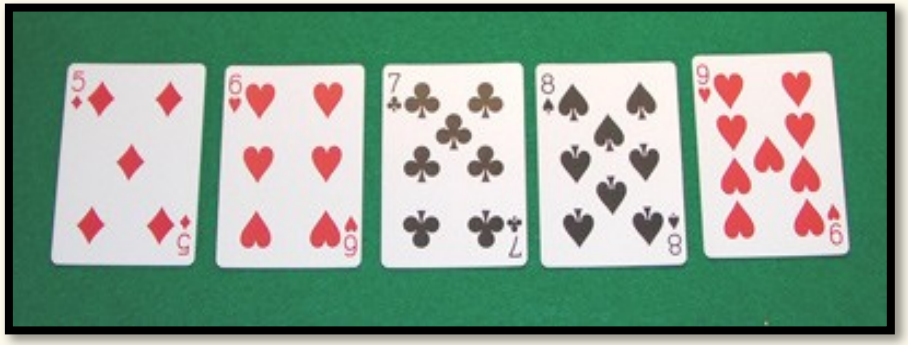
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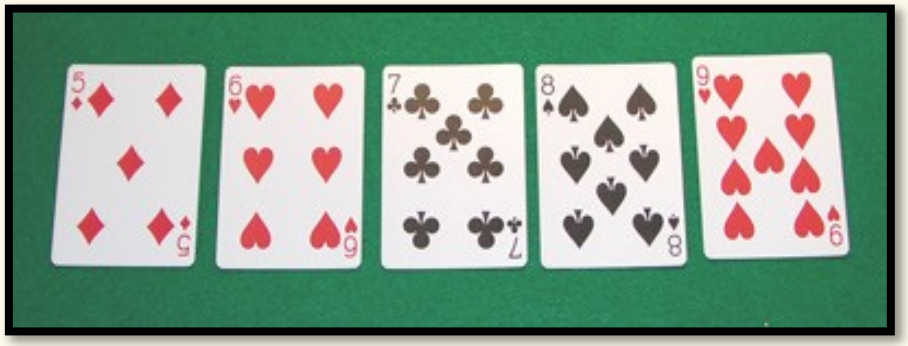
- A "straight" is five consecutive rank cards of any suit (where A,2,3,4,5 also counts as consecutive).
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$$10 \cdot 4^5 = 10{,}240$$

Counting Cards II

- 52 total cards
- 13 different ranks: 2,3,4,5,6,7,8,9,10,J,Q,K,A
- 4 different suits: Hearts, Diamonds, Clubs, Spades

- A flush is five card hand all of the same suit.

How many possible flushes?



Counting Cards II

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How many possible flushes?



$$4 \cdot \binom{13}{5} = 5148$$

Counting Cards III

- 52 total cards
- 13 different ranks: 2,3,4,5,6,7,8,9,10,J,Q,K,A
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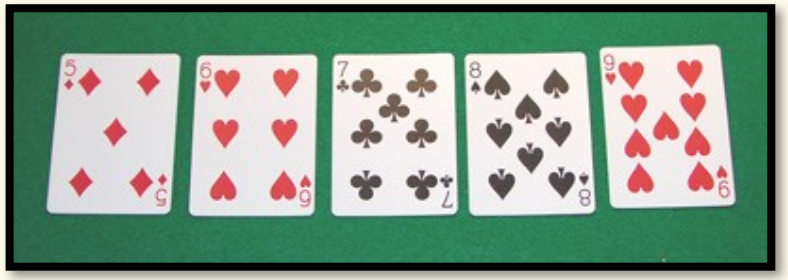
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- How many flushes are NOT straights?



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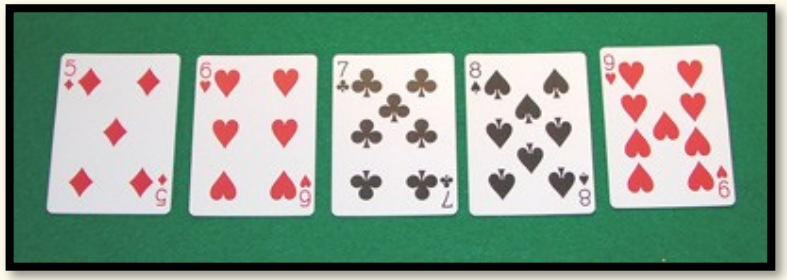
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= #flush - #flush and straight



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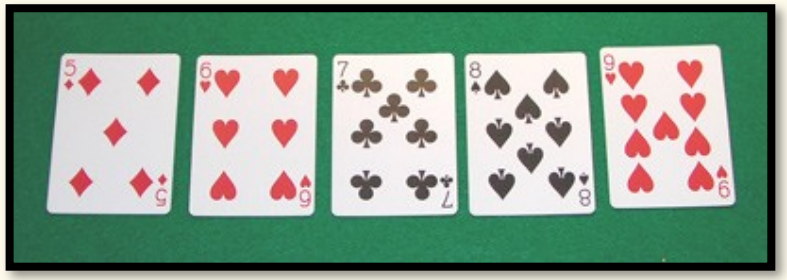
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$$\left(4 \cdot \binom{13}{5} = 5148\right) - 10 \cdot 4$$



Sleuth's Criterion (Rudich)

For each object constructed, it should be possible to reconstruct the **unique** sequence of choices that led to it.

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First choose 3 Aces. Then choose remaining two cards.

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A. Correct
B. Overcount
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https://pollev.com/paulbeame028

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Problem: This counts a hand with all 4 Aces in 4 different ways!

e.g. it counts **A♣, A♦, A♥, A♠, 2♥** ***four*** *times:*

{A♣, A♦, A♥} {A♠, 2♥}

{A♣, A♦, A♠} {A♥, 2♥}

{A♣, A♥, A♠} {A♦, 2♥}

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Use the sum rule

= # 5 card hand containing exactly 3 Aces

+ # 5 card hand containing exactly 4 Aces

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+ # 5 card hand containing exactly 4 Aces $\binom{48}{1}$

Counting when order only *partly* matters

We often want to count # of partly ordered lists:

Let M = # of ways to produce fully ordered lists

P = # of partly ordered lists

N = # of ways to produce corresponding fully ordered list given a partly ordered list

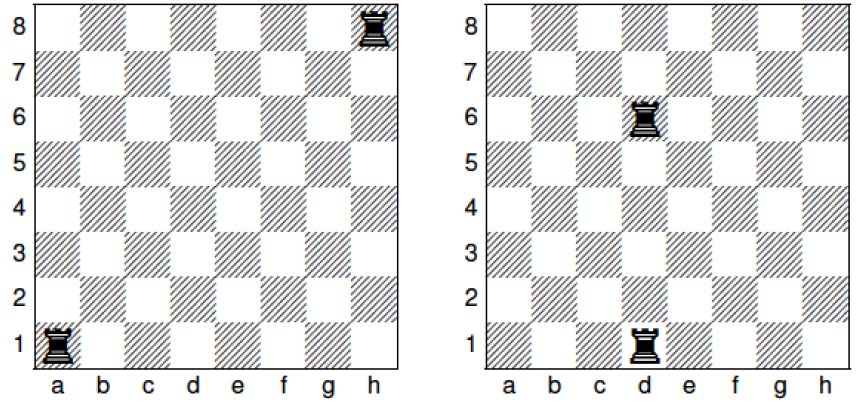
Then $M = P \cdot N$ by the product rule. Often M and N are easy to compute:

$$P = M/N$$

Dividing by N "removes" part of the order.

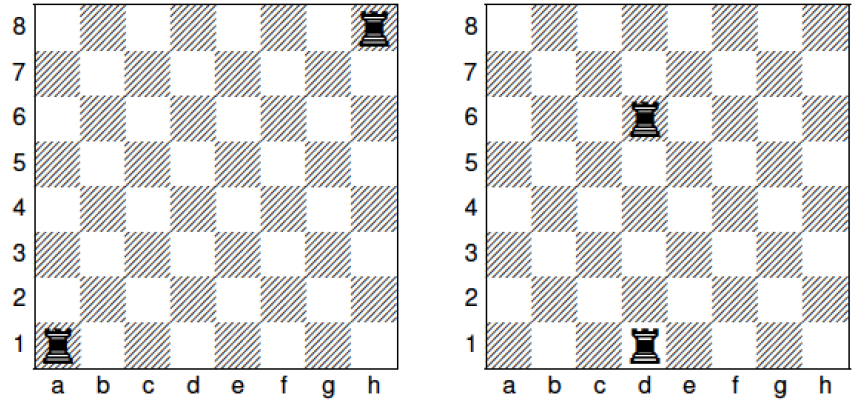
Rooks on chessboard

How many ways to place two identical rooks on a chessboard so that they don’t share a row or a column



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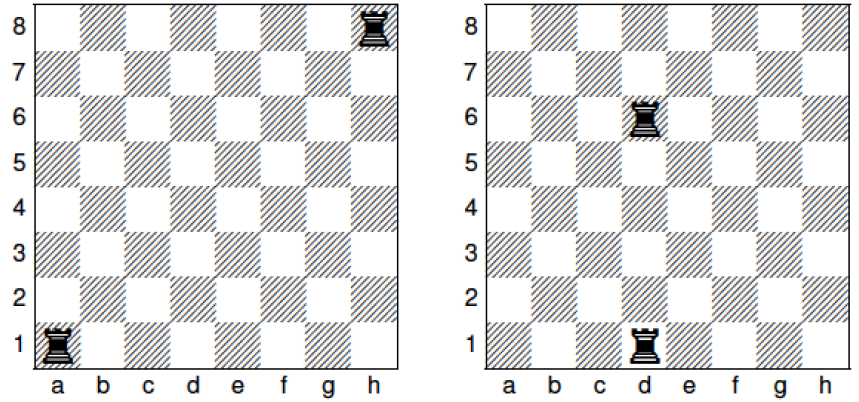


Fully ordered: Pretend Rooks are different

1. Column for rook1
2. Row for rook1
3. Column for rook2
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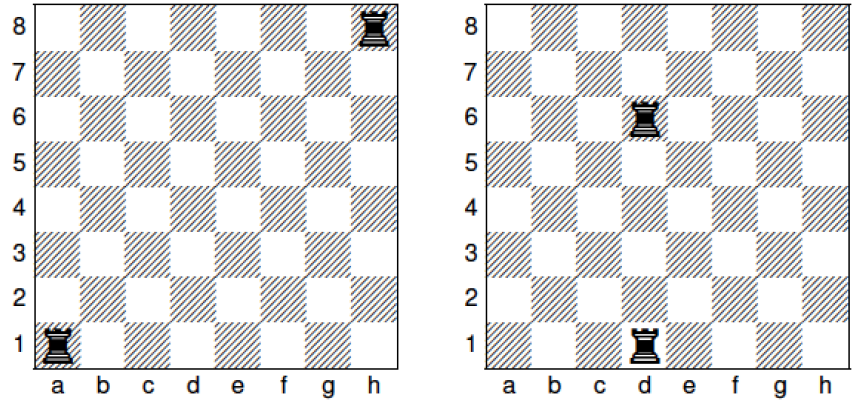
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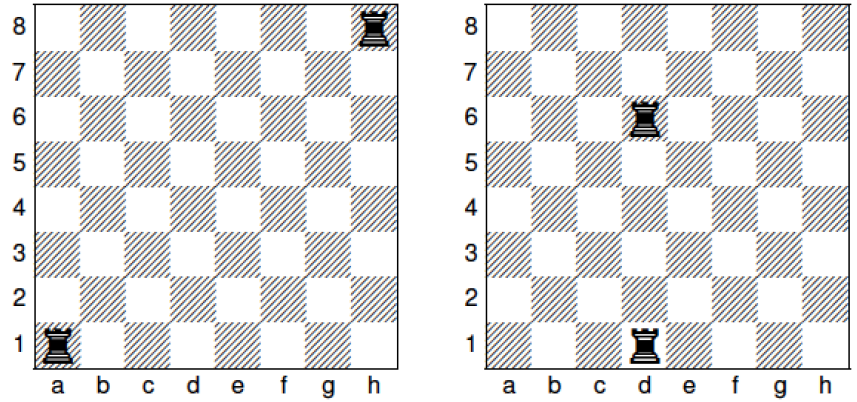
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"Remove" the order of the two rooks:

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(a) valid (b) invalid

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"Remove" the order of the two rooks:

$$(8 \cdot 7)^2/2$$

Binomial Theorem

Theorem. Let $x, y \in \mathbb{R}$ and $n \in \mathbb{N}$ a positive integer. Then,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$$

Set $x = y = 1$

Corollary.

$$\sum_{k=0}^{n} \binom{n}{k} = 2^n$$

Binomial Theorem: A less obvious consequence

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e.g., n=4: 1 4 6 4 1

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$= -1$ if k is odd
$= +1$ if k is even

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Proof: Set $x = -1, y = 1$ in the binomial theorem

Tools and concepts

- Sum rule, Product rule
- Permutations, combinations
- Inclusion-exclusion
- Binomial Theorem
- Combinatorial proofs
- Pigeonhole principle
- Binary encoding/stars and bars